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Conceptualizing thresholds for effective active debris removal in Low Earth Orbit

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The rapid growth of orbital debris in the Low Earth Orbit (LEO) poses an escalating risk to space operations, with existing mitigation proving insufficient to prevent long-term instability. This study models debris population dynamics in the 500–600 km LEO under current FCC 5-year deorbit rules and varying levels of Active Debris Removal (ADR). Using publicly available orbital catalogs and a collision-risk proxy based on object density, cross-section, and relative velocity, simulations depict debris growth and collision risk trajectories under a 30-year period and various scenarios. Results indicate that removal of ~60 large objects (>10 cm) per year is the threshold at which debris growth becomes negative and collision risk declines. This value is scenario-dependent and is presented as an illustrative threshold under controlled assumptions rather than a robust or universal quantitative value. The primary contribution of this study is to demonstrate the existence of a minimum viable ADR regime, which can provide conceptual guidance for debris mitigation policy.

KEYWORDS

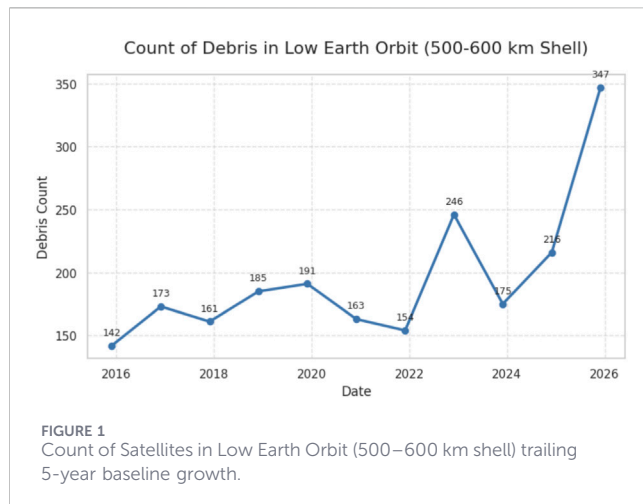
active debris removal (ADR), debris mitigation, FCC (federal communications commission), kessler syndrome, low earth orbit (LEO), policy recommendation

Introduction

The LEO is increasingly crowded with inactive satellites, rocket stages, and other debris, posing a risk to spacecraft operations. According to current estimates ([European Space Agency, 2025b](#)), there are an estimated 1.2 million objects capable of causing damage. Of these, 50,000 large objects are particularly capable of triggering catastrophic fragment-generating collisions through the Kessler Syndrome ([Space.com, 2025](#)). This effect describes a regime in which increasing object density causes self-sustaining collision cascades, severely disrupting existing satellites. Kessler's collision frequency C scales approximately as:

$$C \propto n^2 \sigma v$$

Where n is object number density, σ is effective cross-sectional area, and v is relative velocity. Since collisions scale with n^2 , even small reductions in n will disproportionately reduce collision frequency, creating a nonlinear positive feedback loop ([Kessler and Cour-Palais, 1978](#)). Additionally, larger objects have greater σ , yielding higher collision probability. Thus, debris mitigation policies should aim to remove large objects due to cascading effects on LEO ecosystems. While mitigation efforts in the *status quo*, they are insufficient to prevent long-term instability ([Liou and Johnson, 2006](#)). Long-term risks of orbiting debris in the LEO are projected to increase, which is demonstrated by simulations graphing the Total Number of Objects in Orbit Over Time. Using existing data from NASA Orbital Debris Charts ([NASA Technical Reports Server, 2024](#)), [Figure 1](#) shows that recent years have seen



an exponential increase in debris; this may be attributed to the privatization of space, necessitating active removal.

Recent attempts to remedy this issue, such as the FCC 5-year deorbit rule ([Federal Communications Commission, 2024](#)), have failed to successfully address this issue. As the rule only requires satellites in the LEO to deorbit within 5 years after mission completion, it does not prevent collisions that are resultant of existing large objects. Additionally, it measures time, not risk; treating all post-mission satellites as equally hazardous is misguided, as collision risks differ by object size. Different measures, such as ADR, have emerged as potential solutions, with previous missions such as RemoveDEBRIS ([University of Surrey, Surrey Space Centre, 2025](#)), ClearSpace-1 ([European Space Agency, 2025a](#)), and Astroscale ELSA-d ([Astroscale, 2025](#)) demonstrating improvements in the feasibility of capturing and removing debris. However, such attempts often focus on single missions, without long-term enforcement or accountability upon organizations. At most, these have consisted of technology demonstrations without large-scale action. Evidently, ADR must become more widely applied to meaningfully reduce debris risks. So, this raises a critical question: At what rate will ADR become measurably more effective than mitigation alone?

Unlike previous studies that evaluate ADR feasibility or simulate long-term debris evolution, this study aims to identify whether a nonlinear transition point exists at which ADR qualitatively alters long-term debris dynamics ([Liou, 2011](#)) under an existing regulatory regime. This analysis is intentionally exploratory in nature, and the emphasis is on detecting regime change under policy assumptions, not predicting exact quantitative results. Thus, analysis can reframe ADR as a policy design variable rather than a purely technical variable.

Methods

This study focuses on debris in the 500–600 km range of LEO, which is densely populated and long-lived with respect to orbital decay. Many commercial and scientific missions including Starlink operate in this range, making collisions highly detrimental. Mitigation rules such as the FCC 5-year deorbit rule are designed for this range due to its long-term risk. Objects within this region

experience slow natural deorbiting, making it particularly sensitive to long-term debris growth and high collision risk. Thus, it is exceedingly relevant to study 500–600 km LEO. This analysis is restricted to large objects (>10 cm), as they have the most consequential collisions and generate the majority of debris fragments.

Orbital debris was obtained from publicly available Two-Line Element (TLE) sets from CelesTrak and Space Track archives. Analysis modelled long-term population trends as well as current counts of satellite debris to determine a baseline. To ensure minimal error, analysis removed duplicate objects with incomplete orbital parameters to preserve consistency of datasets. Gemini AI assisted in code generation with manual verification.

Baseline mitigation model

In order to determine a baseline mitigation-only model that assumes the *status quo*, analysis modelled the 30-year projected increase in debris assuming 90 percent global compliance with the FCC 5-year deorbit rule; compliance was considerably high to reflect realistic adherence and allow fair evaluation of ADR effectiveness. New satellite launches were held constant over the simulation period to isolate the effects of the deorbit rule. Because the objective of this study is comparative evaluation, holding launch rates constant ensures that observed differences in debris evolution are solely from removal mechanisms, not fluctuations in launch rates; in real orbital environments, launches are non-linear, which may shift the precise location of the identified threshold but does not alter its existence. Additionally, objects were probabilistically deorbited once they reached the end of their missions and accordingly removed from populations.

Model structure

All parameters, including launch rate, compliance rate, orbital shell bounds, and size cutoff were held constant across all scenarios to ensure that effects were due to ADR intensity.

Debris population was modelled using a discrete-time balance formula:

$$N_{t+1} = N_t + L - D - A$$

Where N_t is the number of large objects (>10 cm) at time t , L represents new launches, D represents post-mission deorbiting under FCC compliance, and A represents ADR removals. Collision-driven growth is captured through the collision-risk proxy instead of explicit fragment generation, which allows for relative comparisons without inaccuracies in precision. In order to illustrate the underlying dynamics that drive threshold behavior, debris evolution can also be expressed in continuous form as:

$$\frac{dN}{dt} = L - D - A + \alpha N^2$$

This continuous formula is included to clarify mechanism rather than imply quantitative analytic precision. Where the term αN^2 reflects the established dependence of collision risk on object density and cross-sectional area; this explains the emergence of a minimum ADR threshold as negative debris growth occurs only when

removals exceed the combined launch inputs and new collision-driven generations.

ADR was implemented as the removal of A objects per year, using various technologies including physical capture, prioritizing the removal of large objects such as defunct satellites and rocket bodies. Removals were applied uniformly throughout each simulation year.

Active debris removal scenarios

To determine the effectiveness of ADR versus limited mitigation measures, this study conducted simulations of 3 policy regimes using monthly data from CelesTrak and Space Track by exporting catalogs for historical debris counts. Graphs were generated using standardized estimates that approximated trends found in the data catalogs.

1. Mitigation only. Post-mission disposal under the FCC 5-year deorbit rule with no ADR.
2. Mitigation with minimal ADR. FCC 5-year deorbit rule compliance was combined with the removal of 10 large objects per year, giving a realistic assessment of current capabilities.
3. Mitigation with scaled ADR. FCC 5-year deorbit rule compliance was combined with increasing ADR removal rates, including removing 20, 30, 40, 50, 60, 70, and 80 objects per year.

Large objects were likely to be out-of-operation spacecraft or rocket bodies or defunct satellite parts, reflecting what current real-world ADR proposals target.

The large range of ADR rates was utilized to probe the full range of possible outcomes, where the lower range (0-10) represented current demonstrated capabilities, with the higher range (20-80) being used to determine whether a meaningful transition in system behavior exists.

Evaluation metrics

This study aims to determine the threshold for when ADR is measurably effective. While many agree ADR has considerable benefits, there is a lack of literature regarding the critical threshold. In order to determine effectiveness, this study uses 3 metrics: total debris population over time, collision risk proxy, and marginal benefit per large object removed, which is defined as the reduction in collision risk per object. ADR is measurably effective when debris growth becomes negative, showing a trend to environmental sustainability and decreased collisions.

In order to maximize controlled settings, all simulations were measured over a 30 year period, where the object populations were frequently updated with satellite de-orbits, ADR, and new launches. Total object count, collision risk proxy, and marginal changes per removed object were calculated at each step.

Collision risk proxy

Collision risk is estimated using a normalized proxy proportional to object density, effective cross-section area, and

relative orbital velocity. For each increase in time, a collision proxy value was computed proportional to the product of these terms, which allowed relative risk to be compared across various simulations. This metric does not represent an absolute collision probability, but rather a relative metric useful for comparing scenarios under controlled conditions. The proxy may be bound by the equation:

$$R = \frac{n \sigma v}{(n \sigma v)_{\text{baseline}}}$$

Where $R = 1$ corresponds to the baseline risk, and $R < 1$ indicates reduced relative collision risk. This baseline assumes the same orbital shell and same population before ADR, which is crucial to avoid false precision and isolate population effects. Absolute collision probability would require Monte Carlo analysis, which is unnecessary for policy comparison. Explicit collision fragmentation modelling was intentionally excluded in order to prevent uncertainty from fragment size distributions and event timings; rather, the collision-risk proxy reflects the nonlinear dependence of collision frequency on the object density and cross-sectional area, ensuring that conclusions retain clarity.

Additionally, the proxy is intentionally simplified and does not capture stochastic collision timing or impact geometry; as a result, it should not be interpreted as an absolute collision probability but rather is designed to preserve the nonlinear dependence of collision frequency on object density and cross-sectional area. Its purpose is comparative rather than predictive; to preserve the nonlinear sensitivity of collision frequency to object density while avoiding false precision. Following, reductions in the proxy should be interpreted as relative directional improvements in collision intensity, not a quantitative result. A decline in the proxy indicated a reduced relative collision intensity between scenarios, not a guaranteed value in realized events.

Limitations

Simulations assume that debris grows at a constant rate, which is utilized to isolate the impact of ADR. These assumptions are intentionally prudent in order to isolate the effects of ADR. More complex dynamics, including fragmentation cascades, would likely increase the required removal rate; thus, the identified threshold is not meant to be universal.

Simulations treated orbital shells as a uniform cloud, which biases against finding a threshold, implying the identified threshold is a lower value. However, this was done in order to avoid introducing unverified assumptions about finescale clustering. While real debris exhibits anisotropies, it would increase local collision rates, which would raise the required ADR rate beyond presented estimates.

Additionally, as all models were run assuming 90 percent compliance to the FCC 5-year deorbit rule, the exact value of the threshold should be treated as an approximate target for policy recommendation, not a universal constant. However, the existence of a minimum ADR is taken directly from nonlinear collision dynamics and is therefore applicable under varying compliance levels.

The absence of a formal sensitivity of uncertainty analysis presents a central limitation of this study. Future, non-linear

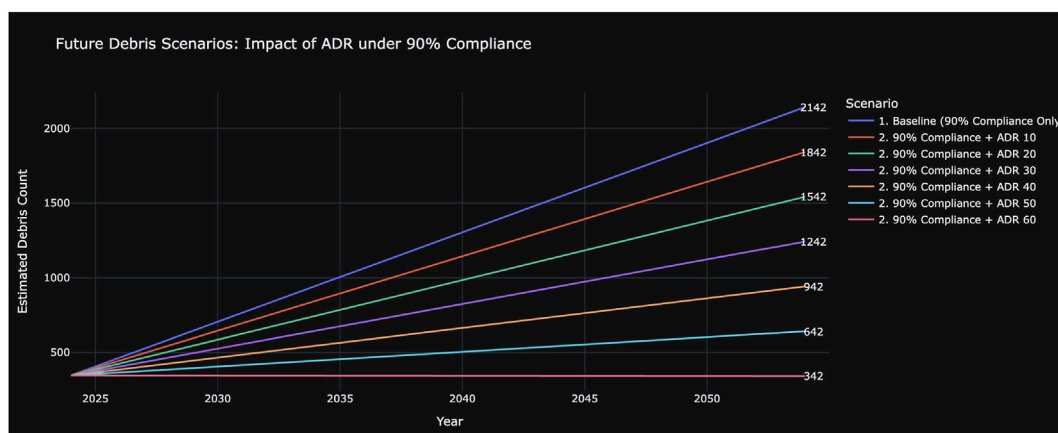


FIGURE 2
Estimated Debris Count vs. Year for baseline and ADR scenarios.

changes in compliance rates, launch growth assumptions, shell width, or size thresholds may impact results. Nevertheless, because the threshold arises from nonlinear collision dynamics instead of fine-tuned parameter choices, its qualitative existence still persists even if the exact numerical location shifts. Thus, this study is meant to be qualitative, not quantitative.

Results

The simulations revealed that while standard FCC 5-year deorbit rules slow the rate of collision risk increase, it will not prevent the debris environment from worsening. Thus, more proactive measures such as ADR are necessary in order to stabilize debris counts in the next 30 years.

Figure 2 compares Estimated Debris Count (LEO 500–600 km) vs. Time. The baseline scenario, assuming 90 percent compliance +0 ADR, saw a 517.3 percent increase in debris count. A total of 2,142 debris (± 30) were estimated by 2055. Each incremental increase in ADR saw a decrease in projected debris counts. The analysis indicates that the threshold for when ADR succeeds in reducing the debris population is removal of ~ 60 objects per year. However, this number is scenario-specific, and a proxy rather than a universal value. This value should be interpreted as an illustrative marker of regime change rather than a precise quantitative requirement. In this scenario, there was a 1.44 percent decrease, meaning that the debris population shrunk, as large objects that were primed for collision were removed. While ADR of 10–50 large objects succeeded in flattening the curve, the total number of debris continued to rise. Thus, these did not stabilize the environment to the degree necessary by this study's metrics. As seen in the scenario assuming ADR of 5 objects, removing greater amounts of debris will incrementally decrease the estimated environmental debris. This places the system further into the region of negative net debris growth, where not only is debris stabilized, but it is also reduced. While this is the ideal scenario for future policy recommendations, it is not

always realistic as ADR is cost-intensive and institutionally complex. This reveals the utility of a “threshold”: if removal of 80–100 large objects is not feasible, it is important to remove at least ~ 60 .

This was supported by Figure 3, which used object density $\times$ cross-section $\times$ relative velocity as a proxy. Under this simulation, the flux metric represents relative collision intensity, where a higher flux correlates to more frequent collisions. In the baseline scenario, flux rate increased linearly. Even assuming compliance as high as 90 percent, the *status quo* will see increasingly difficult operations. Thus, Figure 3 demonstrates that current compliance is not enough, as the risk fails to drop. This is contrasted to ADR, which shows a downwards trend in flux.

Specifically, in scenarios assuming ADR of at least ~ 60 large objects, the collision flux declined significantly, indicating a sustained reduction in collision flux. While the collision risk declined below its baseline value at ADR 70 and 80, the resultant debris growth from projected collisions in ADR 60 is still net negative; thus, positive collision growth can still produce net negative debris due to less colliding objects.

Therefore, the threshold is impactful in multiple metrics; removal of large objects reduces the chances of high-energy collisions, which in turn generate smaller fragments upon impact.

In order to more quantitatively determine the marginal benefit per object removed, simulations analyzed Net Growth Rate vs. ADR Level. Figure 4 depicts the net growth rate of debris in a bar chart, which helps to locate the “tipping point”, or where the bars turn negative.

This point occurred at ~ 60 ADR, where negative numbers represented the net remediation benefit; this was where debris was removed faster than it was added. Following, this backs up that removing at least ~ 60 large objects per year has significant benefits; not only does it stabilize debris growth and reduce collision risks, but it causes a net decrease in added objects per year, actively remediating environments. The identified threshold is supported by trends across multiple metrics, including debris population growth and net growth rate, and is not inferred from the collision proxy alone.

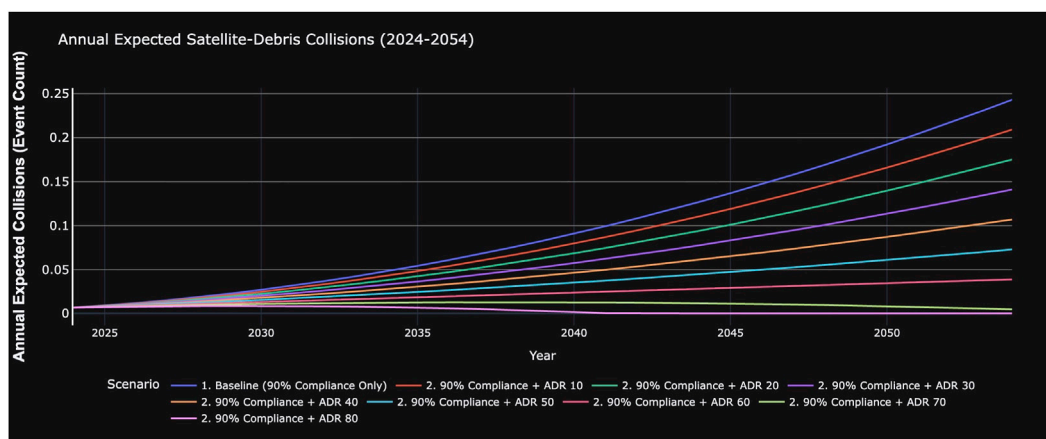


FIGURE 3
Collision flux (collisions/sec for 1 m²) vs. Year. Higher flux corresponds to more frequent collisions.

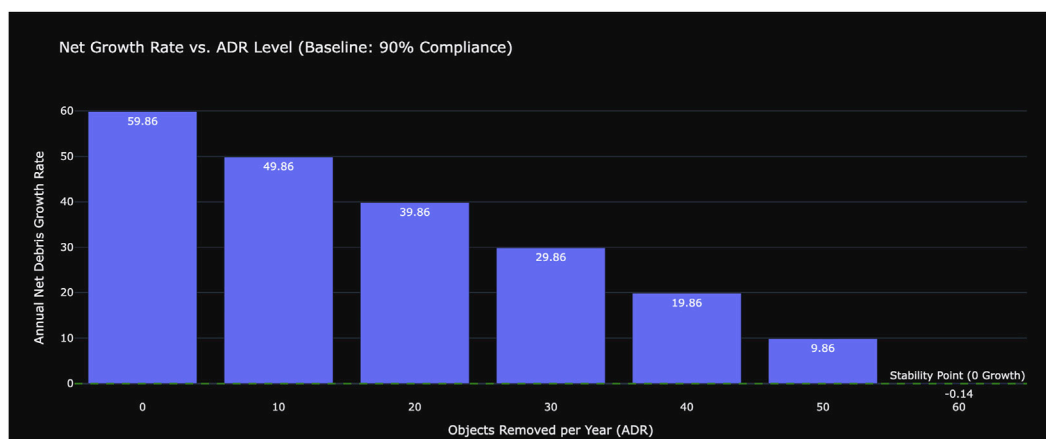


FIGURE 4
Annual Net Debris Growth Rate vs. Objects Removed per Year (ADR). Negative values indicate debris removal exceeds debris generation.

Discussion

Therefore, even near perfect compliance with current mitigation will result in stagnation, not recovery. Failure to shift mitigation efforts will prove detrimental, with debris accumulation becoming increasingly positive; this will magnify the risk of collisions that severely harm existing and future space operations. Thus, it is beneficial for regulators to shift to more proactive measures. This may look like adopting qualitative ADR targets, which sets reasonable goals for future policies. However, it is necessary to clarify that the identified threshold is not proposed as a fixed regulatory mandate, but rather as a reference point that may inform long-term policy design. Policy implications should be framed around shifts in system behavior rather than exact targets. Because the results depend on simplified dynamics and controlled assumptions, the value of this study primarily lies in informing directional policy.

Specifically, policymakers may consider centering removal around ~60 large objects per year, which offers high potential for a trajectory towards sustainable recovery. This precise numerical threshold is contingent on idealized assumptions, but the existence of a minimum viable ADR regime is still a robust implication of nonlinear collision dynamics, making it useful for informing policy. While current technological limitations limit the probability of achieving this exact value, it is important for policymakers to consider the demonstrated potential of ADR. Groundworks laid in RemoveDEBRIS, ClearSpace-1, and Elsa-d support the potential for achieving this threshold, providing a concrete focal point for future mitigation efforts; however, this study proves that limited ADR utilized in these projects is insufficient at its current scope. The potential of substantially increased ADR rates reaching orbital sustainability is evidenced by simulations modelling debris population growth and the collision proxy,

which is a useful indicator of relative risk reduction, but is not intended to justify precise safety claims. In order to guide their recommendations, regimes may consider requiring ADR capacity as a condition for approval of major new launches, which aligns incentives and encourages large-scale adoption. Importantly, this does not necessitate a single actor being held liable for the removal of ~60 objects annually. Rather, it is more realistic for this requirement to be distributed across corporations, countries, and licensing regimes. While ADR remains expensive, the relatively low number of necessitated removals demonstrates feasibility through collective, sustained action. Framing ADR as a collective goal rather than individual mandate reduces widespread resistance and offers potential for long-term sustainability. This study demonstrates that ADR is not only a preventative measure but also may actively reverse long-term instability if properly legislated.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author.

Author contributions

SY: Conceptualization, Investigation, Methodology, Validation, Writing – original draft, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. Authors used Gemini 3.0 to create visual simulations used in my research. They personalized all of the inputs, verified outputs with data from SpaceTrack, and all analyses were their own.

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